

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Deucravacitinib in Participants With Active Systemic Lupus Erythematosus (SLE) **Short Title/Acronym:** POETYK SLE-2 **Protocol Number:** IM011247 **NCT Number:** NCT05620407 **Sponsor Name:** Bristol-Myers Squibb **Investigational Product:** Deucravacitinib (BMS-986165) **Phase of Development:** Phase 3 **Medical Monitor:** Bristol-Myers Squibb Clinical Operations Lead (Coburn Hobar)

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the effectiveness and safety of deucravacitinib compared with placebo in an active moderate to severe Systemic Lupus Erythematosus (SLE) population. **Secondary Objectives:** To evaluate the effect of deucravacitinib on the BILAG-Based Composite Lupus Assessment (BICLA) response, Cutaneous Lupus Erythematosus Disease Area and Severity Index (CLASI-50) response, Lupus Low Disease Activity State (LLDAS), and the ability to taper glucocorticoids.

2.2 Background & Rationale Systemic lupus erythematosus (SLE) is a chronic, progressive, autoimmune disease characterized by heterogeneous clinical manifestations. Deucravacitinib is a first-in-class, oral, selective, allosteric tyrosine kinase 2 (TYK2) inhibitor. Following successful outcomes in the phase 2 PAISLEY SLE trial, this study aims to replicate those findings on a larger scale. It addresses the need for targeted therapies that can provide symptom relief, allow for glucocorticoid tapering, and improve overall quality of life with fewer side effects than current standard-of-care immunosuppressants.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (3:2 ratio for deucravacitinib vs. placebo)

3.2 Planned Enrollment & Duration **Target N\$:** 490 **Number of Sites:** Global (approx. 27 countries across North and South America, Europe, and the Asia-Pacific region) **Estimated Study Start:** 12-Jan-2023 **Estimated Primary Completion:** 19-Oct-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 to ≤ 75 years. 2. Confirmed diagnosis of active moderate to severe Systemic Lupus Erythematosus (SLE). 3. At least one SLE background therapy (immunosuppressant and/or antimalarial) for ≥ 12 weeks before screening, stable for ≥ 8 weeks before screening. 4. If on oral corticosteroids (OCS), dose must be stable for ≥ 2 weeks before screening and cannot exceed 30 mg/day at screening.

4.2 Exclusion Criteria 1. SLE overlap syndromes including, but not limited to, rheumatoid arthritis, scleroderma, and mixed connective tissue disease. 2. Active or unstable lupus neuropsychiatric manifestations, including any condition defined by BILAG A criteria. 3. Active, severe Class III and IV lupus nephritis requiring treatment with cytotoxic agents or high-dose corticosteroids. 4. Known active infection, treatment for chronic infection, or taking more than 1 immunosuppressant at screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Deucravacitinib or matching placebo, twice daily. **Route of Administration:** Oral **Duration of Treatment:** 52 weeks of double-blind treatment, with an optional open-label Long-Term Extension (LTE) period of up to 104 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Background standard-of-care treatment (antimalarials, immunosuppressants). Patients receiving glucocorticoids are instructed to follow a mandatory tapering strategy, unless significant disease activity is present. **Prohibited:** Use of more than one immunosuppressant, parenteral antimicrobial agents within 30 days of randomization, and live virus vaccines.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, baseline SLE disease activity assessment | | **Baseline** | Day 0 | Randomization, First Dose | | **Interim** | Weeks 1 to 51 | 14 scheduled visits for safety labs, disease activity adjudication, and glucocorticoid taper tracking | | **End of Study** | Week 52 | Final Double-Blind efficacy assessment (SRI-4), option to enter 104-week LTE or proceed to Safety Follow-Up | | **Long-Term Extension (LTE)** | Weeks 52 to 156 | 12 scheduled visits for open-label treatment, safety monitoring, and long-term efficacy | | **Safety Follow-Up** | Post-LTE or Early Termination | Final safety assessments (1 visit) |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of patients achieving an SLE Responder Index 4 (SRI-4) response at Week 52. **Measurement Method:** Composite disease activity adjudication assessment (centralized).

7.2 Secondary & Exploratory Endpoints * Proportion of patients achieving BILAG-Based Composite Lupus Assessment (BICLA) response at Week 52. * Proportion of patients achieving Cutaneous Lupus Erythematosus Disease Area and Severity Index (CLASI-50) response. * Achievement of Lupus Low Disease Activity State (LLDAS) and successful glucocorticoid tapering.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via scheduled onsite visits and laboratory tests throughout the 52-week double-blind and 104-week LTE periods. **Serious Adverse Events (SAEs):** Standard reporting timelines to the sponsor within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Utilizes a rigorous management structure for centralized disease activity adjudication and ongoing safety oversight.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy assessments; Safety Set for all patients receiving at least one dose. **Sample Size Justification:** Target enrollment of $N=490$ powered to demonstrate superiority of deucravacitinib over placebo in achieving the primary SRI-4 endpoint, based on phase 2 effect sizes. **Handling of Missing Data:** Non-responder imputation (NRI) for missing binary composite disease activity endpoints (SRI-4, BICLA).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Strict disease activity adjudication monitoring to ensure consistent application of SRI-4 and BICLA criteria across global sites. **Audit Trail:** Standard clinical trial electronic Case Report Form (eCRF) locking and query resolution.

11. Study Identifiers & Governance

Primary ID: IM011247 **Registry Number:** NCT05620407 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** POETYK SLE-2

Official Regulatory Title: A Phase 3, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Deucravacitinib in Participants With Active Systemic Lupus Erythematosus (SLE)

12. Clinical Framework (SLE Specific)

Primary Condition: Systemic Lupus Erythematosus (SLE) **Diagnosis Threshold:** Active moderate to severe disease manifestations (BILAG/SLEDAI criteria) **Medical Methodology:** Selective allosteric TYK2 inhibition superimposed on standard of care **Optimization Target:** Achievement of SRI-4/BICLA response and mandated glucocorticoid tapering

13. Study Procedures & Methodology

Management Pathway: Double-blind experimental treatment overlaid on standard SLE background therapies (immunosuppressants/antimalarials) **Education Protocol (HCP):** Comprehensive disease activity adjudication and glucocorticoid tapering training for investigators **Education Protocol (Patient):** Strict guidelines on adherence to required oral corticosteroid taper **Quality Audit Mechanism:** Centralized adjudication of systemic and cutaneous (CLASI) lupus scoring

14. Observation & Follow-Up Schedule

Total Duration: Up to 156 weeks (52-week double-blind + 104-week open-label extension) **Onsite Visit Cadence:** 14 scheduled visits during the primary 52-week treatment period, plus 12 visits in the LTE, and a final Safety Follow-up visit. **Remote Monitoring Frequency:** As necessary per protocol/patient safety **Checklist Review Interval:** Regular interval tracking of disease activity metrics leading up to the primary Week 52 assessment

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	Dr. Kyriakos A. Kirou	____	06-Apr-2026	Clinical Operations Lead	Coburn Hobar	____	06-Apr-2026
Lead Statistician	TBD	____	06-Apr-2026				